

# Linear algebra – 11

## Non-negative matrices: Perron and Frobenius

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### Theoretical minimum.

**Def.** A matrix  $A \geq 0$  (entry-wise) is *irreducible* if no permutation similarity brings it to a block upper-triangular form; equivalently the digraph on  $\{1, \dots, n\}$  with edges  $i \rightarrow j$  whenever  $a_{ij} > 0$  is strongly connected. It is *primitive* if  $A^m > 0$  for some  $m$ .

**Theorem 1 (Perron 1907).** If  $A > 0$ , then  $\rho(A)$  is a simple eigenvalue with a positive eigenvector, every other eigenvalue has strictly smaller modulus, and the positive eigenvector is unique up to scaling.

**Theorem 2 (Frobenius 1912).** If  $A \geq 0$  is irreducible, then  $\rho(A)$  is a simple eigenvalue with a positive eigenvector, and the eigenvalues of modulus  $\rho(A)$  are exactly  $\rho(A)e^{2\pi i k/h}$ ,  $k = 0, \dots, h-1$ , where  $h$  is the period of the digraph.

**Fact 1.** (Collatz–Wielandt) For  $A \geq 0$  irreducible,  $\rho(A) = \max_{x>0} \min_i \frac{(Ax)_i}{x_i} = \min_{x>0} \max_i \frac{(Ax)_i}{x_i}$ ; in particular  $\min_i \sum_j a_{ij} \leq \rho(A) \leq \max_i \sum_j a_{ij}$ .

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1. Prove Perron's theorem for a strictly positive matrix, and deduce the Collatz–Wielandt formula together with the row-sum bounds for  $\rho(A)$ .
2. (Proposed problems) Let  $N$  be the  $n \times n$  matrix all of whose entries equal  $\frac{1}{n}$  and let  $A \in M_n(\mathbb{R})$  satisfy  $A^k = N$  for some positive integer  $k$ . Show that  $\sum_{1 \leq i, j \leq n} a_{ij}^2 \geq 1$ .
3. (VJIMC 2006, cat. I) Let  $A = (a_{ij})$  be an  $n \times n$  matrix with non-negative entries such that  $\sum_{i,j} a_{ij} = n$ .
  - a) Prove that  $|\det A| \leq 1$ .

- b) If  $|\det A| = 1$  and  $\lambda$  is any complex eigenvalue of  $A$ , show that  $|\lambda| = 1$ .
4. (IMC) For which positive integers  $n$  does there exist an  $n \times n$  matrix  $A$  with entries in  $\{0, 1\}$  such that  $A^2$  is the matrix all of whose entries equal 1?
5. Prove that a non-negative matrix  $A$  is primitive if and only if it is irreducible and aperiodic, and show that then  $A^k/\rho(A)^k$  converges to the rank-one matrix  $uv^T$ , where  $Au = \rho u$ ,  $A^T v = \rho v$  and  $v^T u = 1$ .
6. Let  $A$  be a square matrix all of whose off-diagonal entries are positive. Prove that the right-most eigenvalue of  $A$  in the complex plane is real and that all the other eigenvalues lie strictly to the left of it.